Real-World Effectiveness Of Dapagliflozin And Sitagliptin Fixed Dose Combination In Patients With Type 2 Diabetes Mellitus In India: Impact On Blood Glucose Levels In A Retrospective Study From Electronic Medical Records

*Prakadeesh Bharathi[1], Vinay Kudrigikar[1], Prachi Ahire[2], Unnikrishnan A G[3], Shruti Dharmadhikari[1], Chintan Khandhedia[1], Neeraj Markandeywar[1], Amey Mane[1], Suyog Mehta[1]

[1]Sun Pharma Laboratories Limited, Mumbai, India
[2]Sun Pharmaceutical Industries Limited, Mumbai, India
[3]Chellaram Hospital-Diabetes Care & Multispeciality, Pune, India

*Presenting Author

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Rationale of Study

Real-world data are important to understand the generalizability and applicability of FDC of dapagliflozin and sitagliptin, in clinical practice and to help inform everyday patient treatment decisions for clinicians.

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To date, there is limited real-world data on effectiveness of FDC of dapagliflozin and sitagliptin in India.

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This retrospective, observational, EMR-based study aims to assess the real-world effectiveness and usage pattern of FDC of dapagliflozin and sitagliptin in Indian patients.

Objective and Study Design

SUN PHARMA

To evaluate the effectiveness of dapagliflozin and sitagliptin fixed dose combination in patients with type 2 diabetes mellitus in India

Study Design:

- ➢ This is a real-world, retrospective, observational, electronic medical record (EMR) based study.
- ➢ In this study, EMR data from the patients satisfying the eligibility criteria was retrieved and analyzed

Inclusion Criteria:

- Male and female patients of age ≥18 years.
- Patients diagnosed with type-2 diabetes mellitus.
- Patients with HbA1c ≥7% at baseline visit.
- Patients prescribed dapagliflozin and sitagliptin FDC in any visit other than baseline visit on the EMR platform.
- Patients with EMR data captured for more than one visit.

Exclusion Criteria:

- Patients diagnosed with type-1 diabetes mellitus.
- Patients are on insulin or any other injectable antidiabetic medication like glucagon-like peptide-1 (GLP -1) agonists.
- Patients with incomplete medical records with respect to the required study parameters.

Methodology: Study Design

The study retrieved EMRs of patients who met the eligibility criteria and had been prescribed dapagliflozin and sitagliptin fixed-dose combination (FDC) and were at least 18 years old.

Retrieve EMR data → Filter patients meeting eligibility criteria → Visit 1 (Baseline visit) → Visit 2 (3 months ± 1 month from baseline) → Visit 3 (6 months ± 1 month from baseline)

Demographic characteristics, symptoms, diagnosis, HbA1c, glucose levels, comorbidities, medication details

- **The Baseline visit (Visit 1) reading was the visit in which the patient was prescribed with dapagliflozin and sitagliptin FDC as captured in EMR (2022-2023)**
- **And at follow-up visits (Visit 2 and Visit 3) reading after the baseline visit in this study.**

Study Endpoints

Primary Endpoint:

- Mean change in HbA1c from baseline to 3 months in patients with HbA1c ≥8% at baseline.

Secondary Endpoint:

1. Mean change in FBG level from baseline to 3 months and 6 months in patients with HbA1c ≥8% at baseline.
2. Mean change in PPBG level from baseline to 3 months and 6 months in patients with HbA1c >8% at baseline.

Results: PATIENT FLOW

Patient Disposition (Category)	Patient count (n)
Total patients in EMR records from 2022-2023	5213919
Total patients in EMR ≥18 years	4667883
Patients in EMR ≥18 years diagnosed with type 2 diabetes mellitus	1199314
EMR records of patients with Dapagliflozin and Sitagliptin fixed-dose combination (FDC)	58020
EMR records of patients prescribed with Dapagliflozin and Sitagliptin fixed-dose combination (FDC) in any visit other than the baseline visit on EMR platform.	32130
Patients with glycosylated hemoglobin (HbA1c) ≥8% at baseline visit prescribed with dapagliflozin + sitagliptin FDC in any visit other than the baseline visit on EMR platform	8525
Patients with glycosylated hemoglobin (HbA1c) ≥7% at baseline visit prescribed with dapagliflozin + sitagliptin FDC in any visit other than the baseline visit on EMR platform	12060
Patients meeting eligibility (inclusion and exclusion) criteria	**3112**

Demographic and Other Baseline Characteristics

Parameter	Category	Patient Count (n) & % N=3112	Mean Value
Age (Years)		3112	54.71 ± 10.6
Gender	Male	1699 (54.5 %)	
	Female	1413 (46.5 %)	
Weight (Kg)			72.77 ± 12.33
BMI (kg/m2)	Overall	1690	28.24 ± 4.98
	Underweight (<18.50 kg/m2)	9 (0.5 %)	17.76 ± 0.67
	Normal (18.50 - 22.99 kg/m2)	161 (9.5 %)	21.72 ± 1.1
	Overweight (23.00 - 24.99 kg/m2)	235 (13.5%)	24.08 ± 0.59
	Obese (>= 25.00 kg/m2)	1285 (76.5%)	29.9 ± 4.52

Results: Mean change in HbA1c

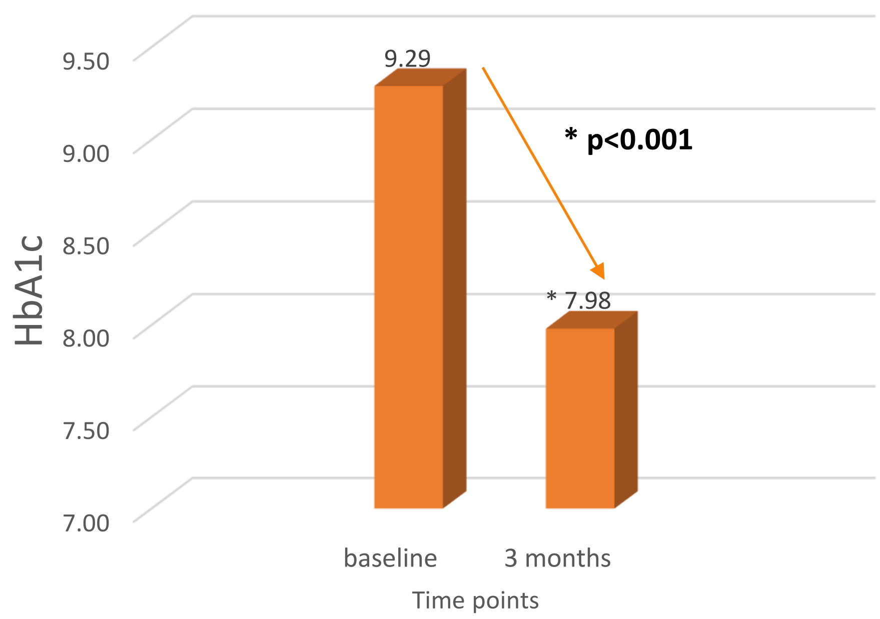
Results: Primary end point

Of 3,112 EMRs included, 838 were eligible for primary endpoint assessment.

Mean HbA1c at baseline was 9.29% which reduced to 7.98% at 3 months

(change from baseline [CFB] -1.31; p<0.001)

Mean change in HbA1c from baseline to 3 months in patients with HbA1c ≥8% at baseline



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Results: Change in FBG

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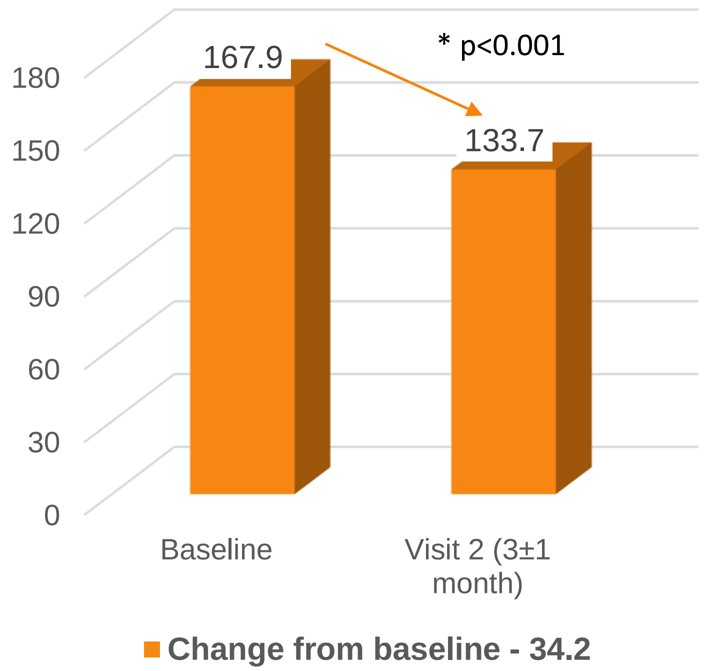
For 710 patients, FBG data was available for baseline and Visit-2 (3±1month).

Change from baseline (CFB) at Visit-2 was statistically significant (167.9±55.3 to 133.7±40.0)

CFB: -34.2±57.8; p<0.0001.

n = 710

Change in FBG: at 3 months



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Results: Change in FBG

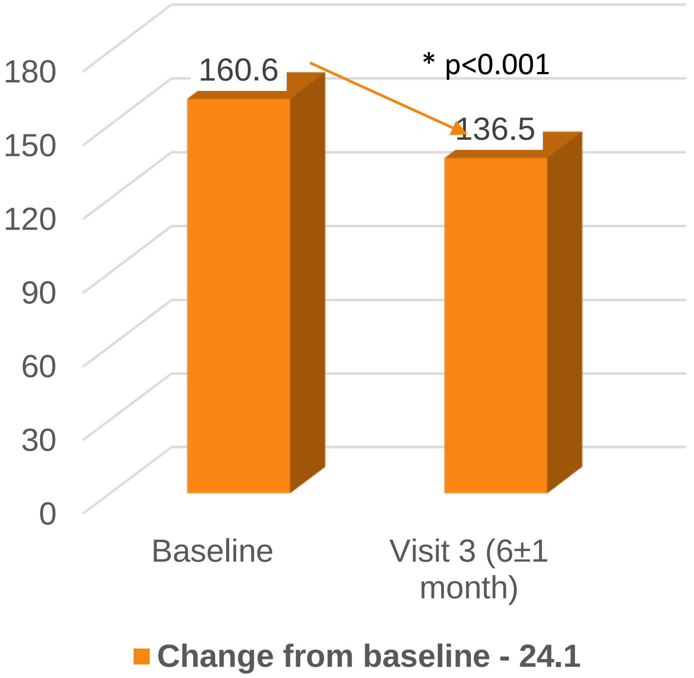
For 323 patients, FBG data was available for baseline and at Visit-3 (6±1 month)

CFB at Visit-3 was statistically significant (160.6±47.8 to 136.5 to 42.7)

CFB -24.1±55.6, p<0.0001.

n = 323

Change in FBG: at 6 months



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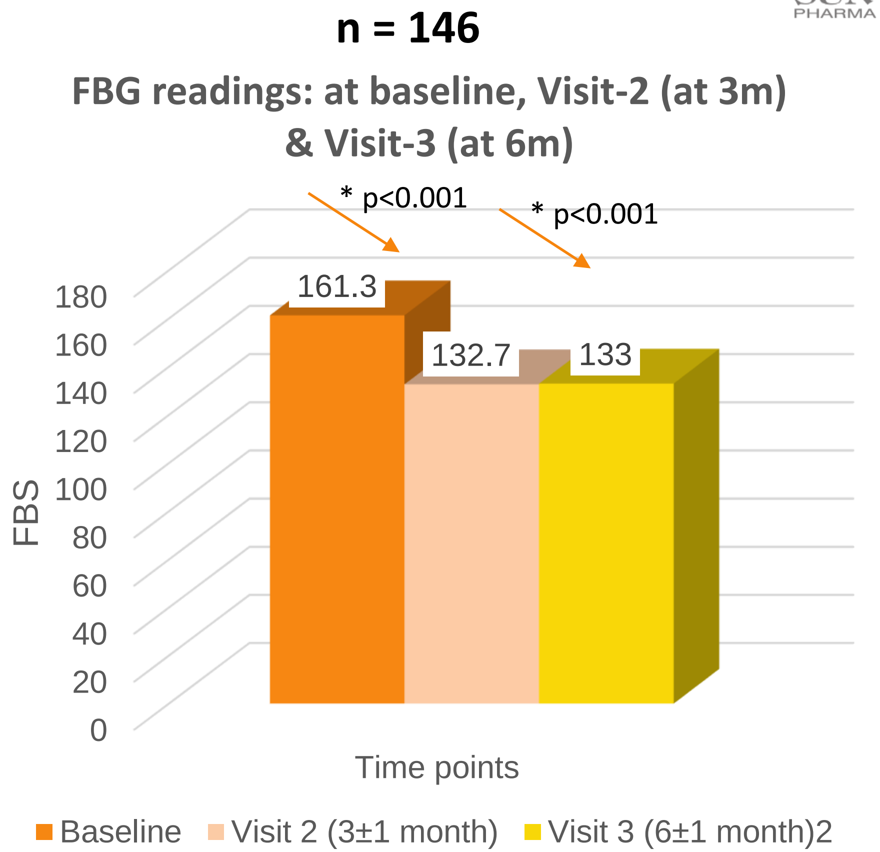
Results: Change in FBG

For 146 patients, FBG readings were available at baseline, Visit-2 and Visit-3

CFB at Visit-2 (161.3±46.2 to 132.7±37.5; **CFB: -28.56±51.8; p<0.0001**)

Visit-3 (161.3±46.2 to 133.0±37.7; **CFB: -28.3±50.5; p<0.0001) was statistically significant**



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Results: Change in PPBG

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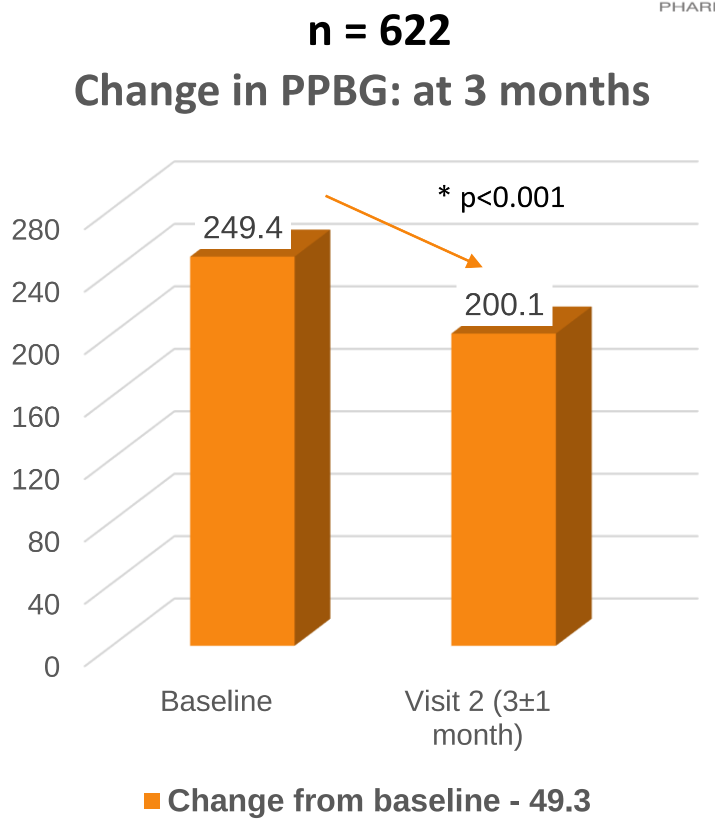
For 622 patients, PPBG data was available for baseline and Visit-2 (3±1month).

Change from baseline (CFB) at Visit-2 was statistically significant (249.4±72.1 to 200.1±62.38)

CFB=-49.3±80.2; p<0.0001)



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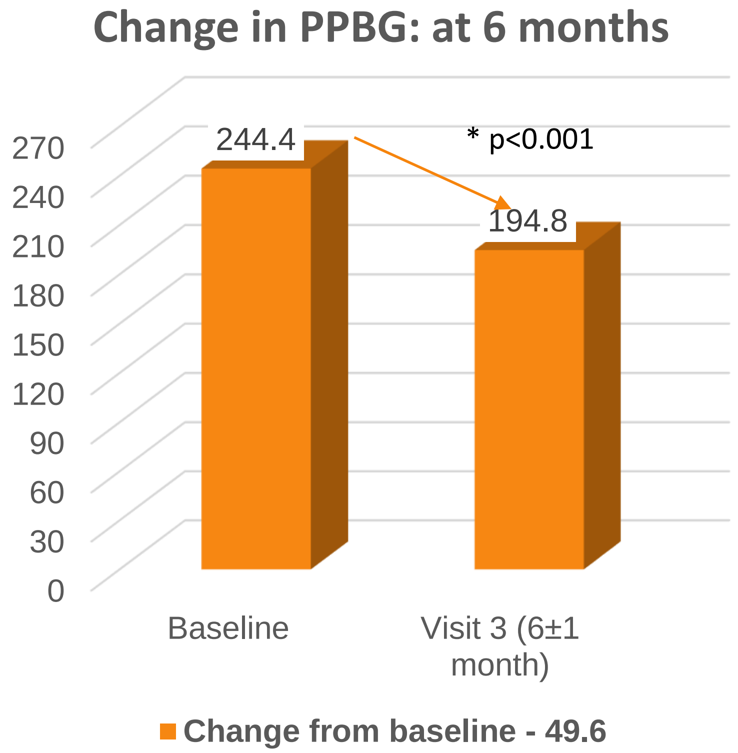
Results: Change in PPBG

For 264 patients, PPBG data was available for baseline and at Visit-3 (6±1 month

CFB at Visit-3 was statistically significant (244.4±67.6 to 194.8±61.5)

CFB=-49.6±77.7; p<0.0001)

n = 264



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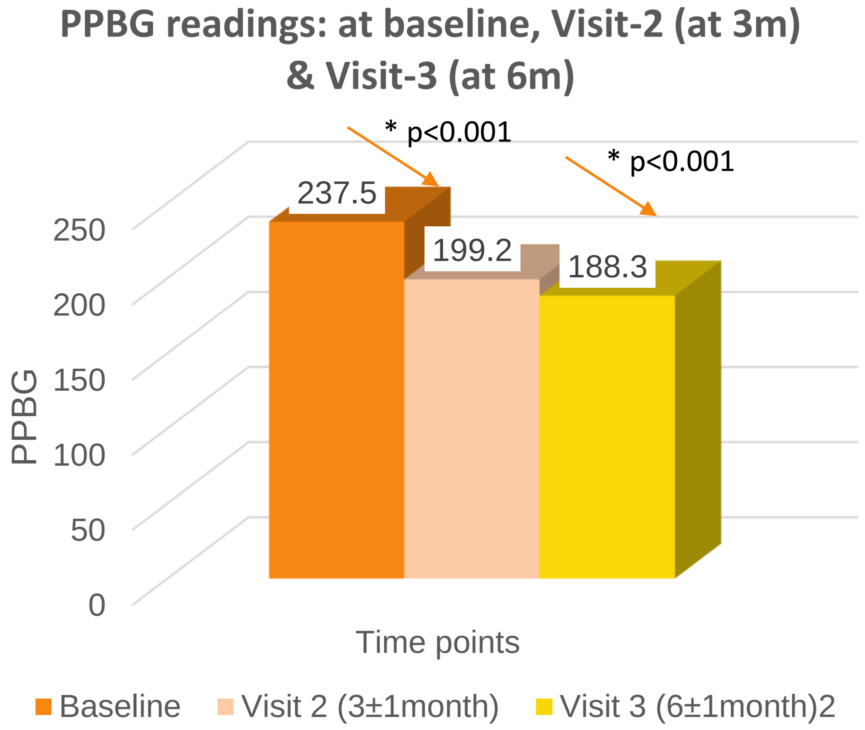
Results: Change in PPBG

n = 123

For 123 patients, PPBG readings were available at baseline, Visit-2 and Visit-3

- CFB at Visit-2 (237.5±65.5 to 199.2±9.0; **CFB: -38.3±80.7; p<0.0001)**
- Visit-3 (237.5±65.5 to 188.3±54.6; **CFB: -49.2±71.0; p<0.0001) was statistically significant.**



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Study Conclusion

- This retrospective, EMR-based study demonstrated effectiveness of dapagliflozin-sitagliptin FDC, in patients with T2DM and HbA1c≥8%, with respect to significant improvement in FBG and PPBG levels.

Thank you